Lab 7: Regression Discontinuity Design

PHS2000B, Spring 2025

Nicky Rahim

April 02, 2026

Overview

1. **Intuition behind RDD**
2. **RDD features**
3. **Sharp vs Fuzzy**
4. **Effect Types**
5. **Excersize**

Intuition behind RDD

Discontinuities in a Continuous World

We generally expect things to be continuous or gradual in nature

But we often create various thresholds for ease of treatment assignment or for cost-effectiveness.

Discontinuities in a Continuous World

Some examples:

- Having thresholds make decisions easier for medical providers
  - Low birth weight
  - Advanced maternal age
  - Lab test ranges, thresholds for clinical risk scores
- Thresholds as a form of austerity
  - Pension benefits for retirees
  - Allocation of scarce medications (BMI thresholds for COVID vaccines)

Any that come to mind from your fields?

RDD features

When can we use regression discontinuity?

- There is a continuous variable that is used to determine eligibility for treatment. In RDD we will call this the **running (or "forcing") variable**. Examples: test scores, GPA, birth weight, income, CD4 count, maternal age, vote share
- **There is a cut-off or threshold** in the running variable, at which point the probability of treatment or exposure changes discontinuously (it "jumps"). The location of this cut-off can be somewhat arbitrary, but above it, the probability of treatment is different than the probability below it.
- There is a **bandwidth** around the cutoff for which there are still observations
- **Continuity assumption**: counterfactuals are continuous functions of X across the threshold c ($lim_{x \to c^-} \; E[Y^a|X] = lim_{x \to c^+} \; E[Y^a|X]$). Do we think this is testable?

Why do folks like regression discontinuity designs?

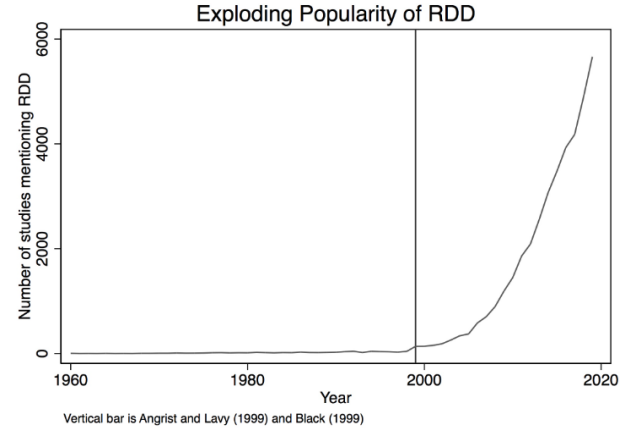
1. **They reflect real world treatment assignment mechanisms**

RDDs are one of the few study designs where an investigator knows the treatment assignment mechanism. Knowing the assignment mechanism (or "rule") allows us to develop models which more accurately capture the "selection mechanism", i.e., the mechanism that leads to certain people being treated while others not.

2. **They have high internal validity**

Studies have shown that RDD estimates have very high internal validity. For example, Chaplin et. al. (2018) test estimates from 15 RDD studies against comparable randomized controlled trials and find that the RDD studies very closely approximate the results from the RCTs.

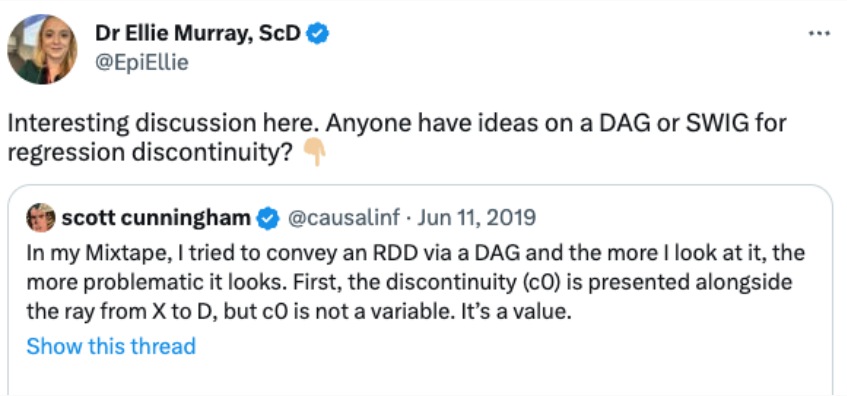
RDD's popularity



Cunningham, Scott. Causal inference: The Mixtape. Yale University Press, 2021.

At this point you are probably dying for a DAG

We love DAGs. But sometimes, our love for DAGs is unrequited :(



Sharp vs Fuzzy RDD

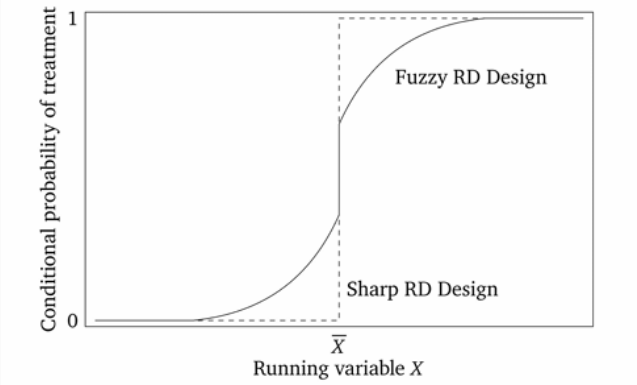
Sharp vs Fuzzy

Sharp RDD

- A sharp cutoff where everyone on one side is treated and no one on the other side is not (dashed line)
-
$P(A = 1|X > c) = 1;\ \ P(A = 1|X < c) = 0$

Fuzzy RDD

- A fuzzy cutoff where probability of treatment changes discretely at the cutoff, but there is imperfect compliance (solid line)
- $P(A = 1|X > c) > P(A = 1|X < c)$



Can we think of examples of each?

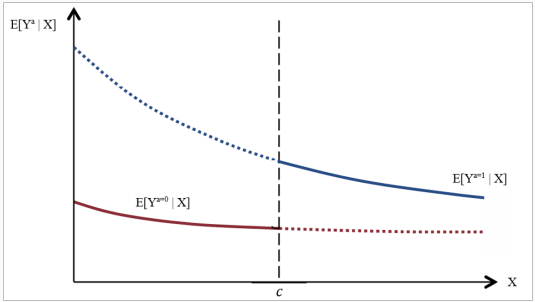
Challenge of causal inference in a sharp RDD

Let $A = \{0, 1\}$ be a binary treatment. $A = 1$ when $X \geq c$. That is, c is the threshold.

$E[Y^a|X]$ is the expected value of the outcome Y when treatment is set to a, given a specific value of X.

Solid line = observed potential outcomes
Dashed line = unobserved potential outcomes

We can only observe potential outcomes under treatment when $X \geq c$ and potential outcomes under no treatment when $X < c$. In other words, sharp RDDs suffer from a violation of structural positivity.

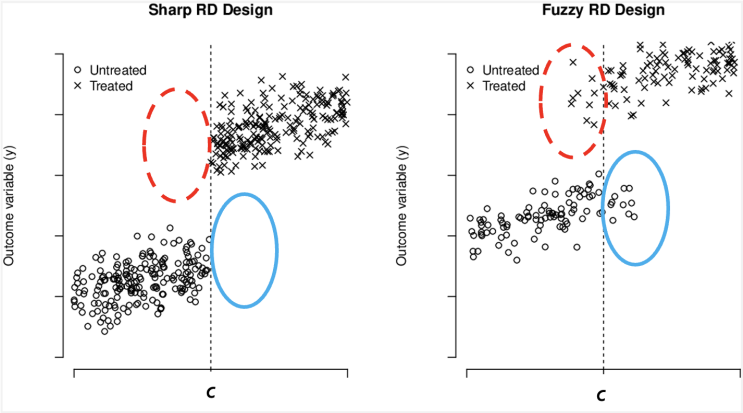


Adapted from Bor et al. (2014)

Challenge of causal inference in a fuzzy RDD

Fuzzy RDDs can be thought of as sharp RDDs with **imperfect compliance or adherence**

Imperfect compliance might make us worry about why some individuals with $X < c$ are receiving treatment and why some individuals with $X \geq c$ are not



Fuzzy RDD

- We have an arbitrary threshold that is strongly (but not perfectly) determines treatment status
- We assume that the arbitrariness of the threshold means that there aren't other changes at the threshold that can affect the outcome
- This is exactly like an instrumental variable situation!

Estimating causal effects in sharp vs fuzzy RDD

Sharp RDD

We are usually interested in estimating the **Average Causal Effect (ACE)** at the threshold

$$ACE_{SRD} = E[Y_i^{a=1} - Y_i^{a=0} | X = c]$$

Using our data:

$$ACE_{SRD} = \lim_{x \to c^+} E[Y_i | X_i = x] - \lim_{x \to c^-} E[Y_i | X_i = x]$$

Fuzzy RDD

We are usually interested in estimating the **Complier Average Causal Effect (CACE)** at the threshold

$$CACE_{FRD} = E[Y_i^{a=1} - Y_i^{a=0} | X = c, \text{compliers}]$$

Using our data:

$$CACE_{FRD} = \frac{\lim_{x \to c^+} E[Y_i | X_i = x] - \lim_{x \to c^-} E[Y_i | X_i = x]}{\lim_{x \to c^+} E[A_i | X_i = x] - \lim_{x \to c^-} E[A_i | X_i = x]}$$

Estimating causal effects in sharp vs fuzzy RDD

Sharp RDD Model

$$Y_i = f(X_i) + \beta A_i + \epsilon_i$$

Fuzzy RDD Model

Same as IV, estimate two-stage least squares (2SLS) model ($Y_i = f(X_i) + \beta \hat{A}_i + \epsilon_i$)

Identifying Assumptions in terms we know

Assumption	Sharp RD design	Fuzzy RD design
Relevance	By definition	Needed (check for discontinuity in treatment at threshold)
Independence	People cannot manipulate the running variable + the threshold cannot coincide with a jump in another variable that affects the potential outcome	
Exclusion	The threshold cannot cause a jump in another variable which affects the potential outcome	
Monotonicity	Not needed (everyone is a complier)	Needed (just like IV)

'Independence' and 'Exclusion' pertain to our assumption of continuity of counterfactuals at the threshold.

Effect Types

Causal effects (so far)

Average Treatment Effect (ATE) is the effect of treatment in the whole population. $E[Y_i^{a=1} - Y_i^{a=0}]$

Average Causal Effect (ACE) is the effect of treatment at the threshold. $E[Y_i^{a=1} - Y_i^{a=0}|X = c]$. ACE = ATE if we assume that the effect at the threshold applies to everyone (very very unlikely)

Local Average Treatment Effect (LATE) is the effect of treatment among compliers. $E[Y_i^{a=1} - Y_i^{a=0}|\text{compliers}]$

Complier Average Causal Effect (CACE) is the effect of treatment among compliers at the threshold. $E[Y_i^{a=1} - Y_i^{a=0}|X = c, \text{compliers}]$

Effect of the threshold / ITT at the threshold is the effect of the threshold mechanism. $E[Y_i^{z=1} - Y_i^{z=0}|X = c]$, where z is an indicator for being on one side of the threshold.

Threats to exclusion

There are many settings where we worry that being designated as above/below a threshold can have an effect on outcome of interest through avenues besides the treatment.

In these cases, econometricians tend to 'reframe' the research question to evaluate the effect of the designation at the the threshold (ITT at the threshold).

RDD in practice

Estimation

1. **Center the running variable around the cut-off**

The value 0 in the centered data will correspond to the threshold.

2. **Create a "discontinuity sample" by trimming your sample to include observations within a certain bandwidth h around the cut-off**

Sharp RDD

- Fit the regression: $Y_i = f(X_i) + \delta A_i + \epsilon_i$
  - $f(X_i)$ is the functional form imposed on the running variable (i.e linear, quadratic...)
  - δ_i is the coefficient of interest on our treatment A

Fuzzy RDD

- Fit a 2SLS:$A_i = f(X_i) + \gamma \mathbb{I}(X > c) + \mu_i$ & $Y_i = f(X_i) + \delta \hat{A}_i + \epsilon_i$
- δ_i is the coefficient of interest on our treatment A

Selecting bandwidth

Selecting the bandwidth, like many things in statistics, is a science and an art

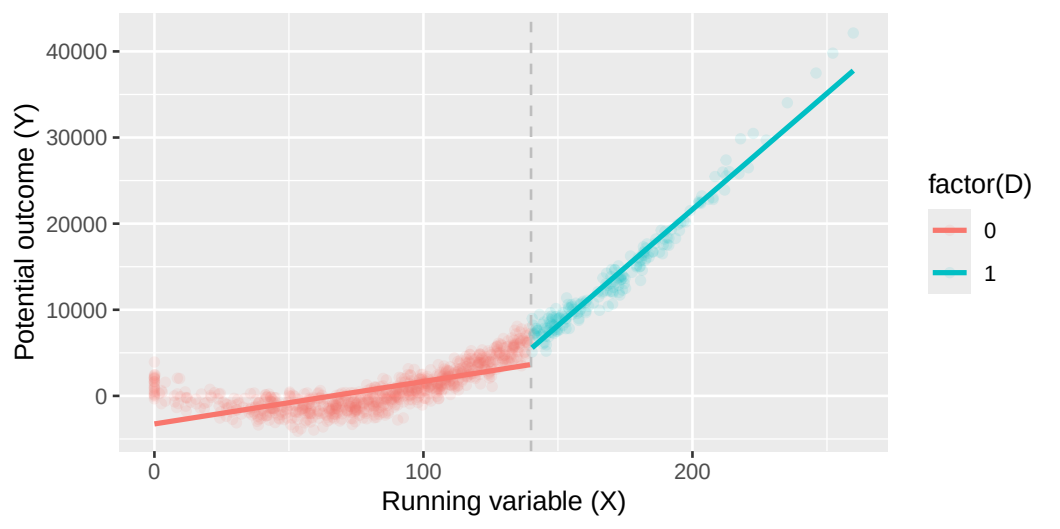
Things to keep in mind:

- There is an inherent tradeoff: for bandwidth $c - h \leq X \leq c + h$, as h gets smaller, precision goes down, but bias in estimates are reduced.
- There are data-driven methods to select the bandwidth that minimizes the mean squared error
- Run sensitivity analyses with a multiple bandwidths to see how estimates change

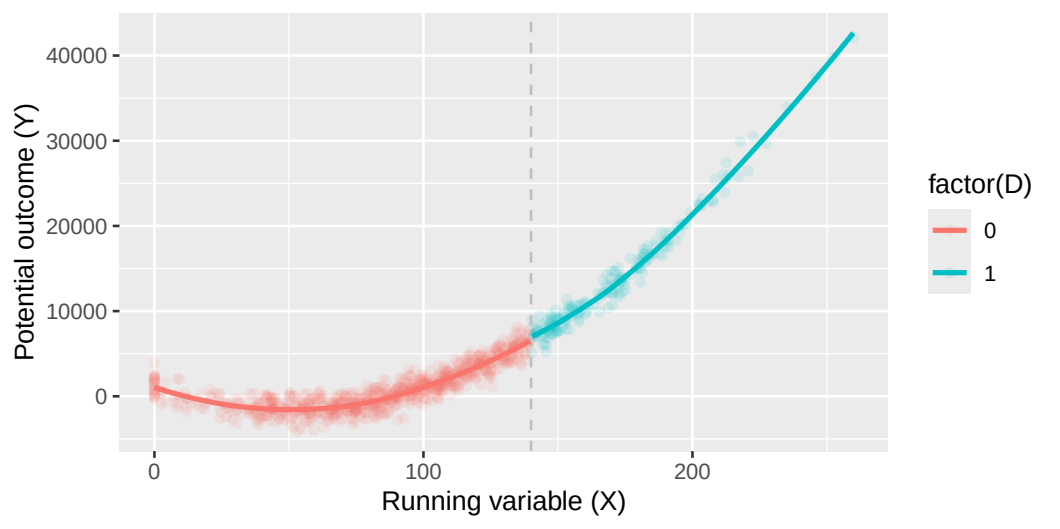
Functional form choices

- You should also do sensitivity analyses related to the functional form specifications used in the RDD model as well as choice of bandwidth.
- Gelman and Imbens (2019) warn against using higher-order polynomial models
  - Hard to choose what order of the polynomial is optimal in a systematized way linked to causal inference as opposed to goodness of fit
  - When inspecting the weights implied by higher order polynomials can take on extreme values (overweighting certain data points)
  - Confidence intervals can fail to include zero with probability higher than the type I error rate
- Can be circumvented via local-linear/non-parametric estimation

Fucntional form choices



Functional form choices



Diagnostics

1. **Balance checks** Visually explore relationship of covariates with running variable. Fit your RDD model, except have the covariate as the outcome, to empirically check for balance of measured covariates.
2. **Placebo Thresholds**

Empirically test for jumps in the outcome along the running variable by splitting the dataset at the cut-off and, in each dataset, estimating the RDD model at "placebo thresholds"

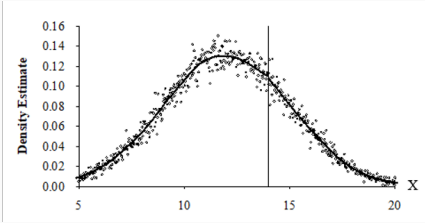
Diagnostics

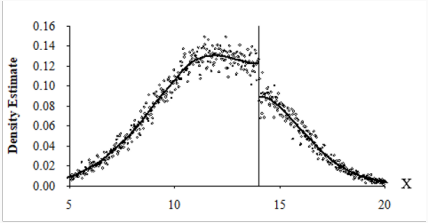
3. **Visually check for sorting around the threshold for the running variable**

Sorting or "bunching" around the cut-off may indicate manipulation of the running variable. This violates the continuity assumption

We can visually inspect for bunching by creating a histogram/density plot of the running variable and assessing if there are discontinuous changes in the density at the cut-off

We can also conduct a McCrary Density Test, which treats frequency counts in histogram as dependent variable in a local linear regression (H_0: density does *not* increase at the threshold)





Data Excersize

CD4 count

CD4 counts measure the number of T lymphocyte cells per cubic millimeter of blood, measure of severity of HIV infection (low CD4 = worse infection).

When ART was first developed, it was often rationed based on CD4 count as eligibility to ensure the sickest received treatment. We are interested in studying the effect of initiating treatment on ART on body weight.

If we tried to naively estimate the effect of treatment initiation on our body weight outcome, where would we run into problems? We can use CD4 count treatment eligibility (< 350) for an RDD analysis (in fact, this has been done to estimate the effects of treatment on mortality in Bor et al 2014)

CD4 count

- **cd4_lab**: Lab measured CD4 count (running variable)
- **art**: Indicator variable for ART receipt (treatment)
- **Y**: Weight (kg) 12 months after CD4 measurement (outcome)
- **age**: Age measured in years (covariate)
- **income**: Income in ZAR per month

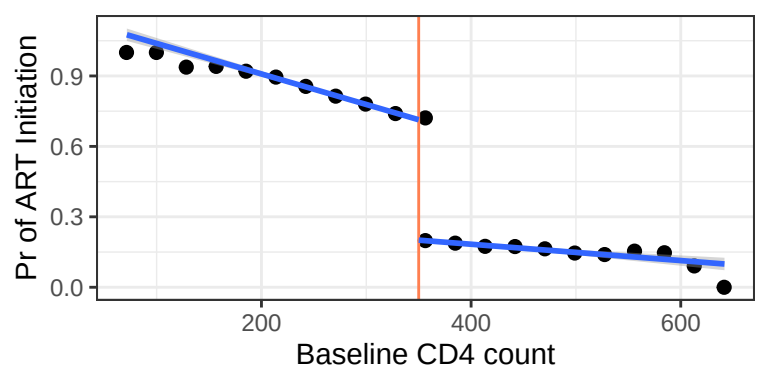
CD4 count: data preparation

```r
load("RDdata.Rda")
rd.data$cd4_centered <- rd.data$cd4_lab - 350
rd.data$elig <- ifelse(rd.data$cd4_centered < 0, 1, 0)
head(rd.data)
```

```
##    cd4_lab art        Y      age   income cd4_centered elig
## 1 104.0670   1 35.74248 24.25241 11157.97    -245.9330    1
## 2 123.3415   1 88.40221 28.39326 15943.60    -226.6585    1
## 3 112.6995   1 45.32758 44.88341 11031.83    -237.3005    1
## 4 122.2668   1 58.13979 32.27773 13964.77    -227.7332    1
## 5 100.5393   1 52.43089 16.85430 10265.47    -249.4607    1
## 6 117.8136   1 61.74176 16.47074 20194.80    -232.1864    1
```

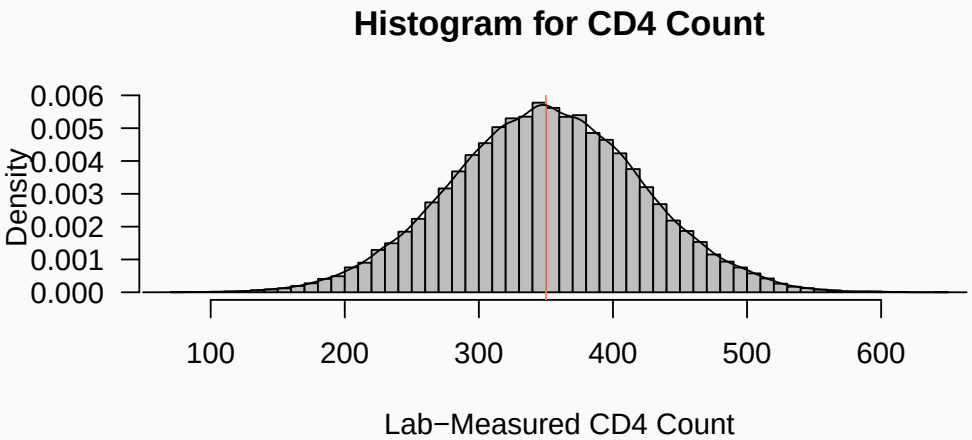
CD4 count: sharp vs fuzzy?

```
ggplot(rd.data, aes(x=cd4_lab, y=art, group=elig)) +
  stat_summary_bin(fun.y='mean', bins=20, size=2, geom='point') +
  geom_vline(xintercept = 350, color = "coral") +
  geom_smooth(method='lm', formula=y~x) +
  labs(x = "Baseline CD4 count", y = "Pr of ART Initiation") +
  theme_bw()
```



CD4 count: visual evidence of bunching?

```
hist(rd.data$cd4_lab,main="Histogram for CD4 Count",xlab="Lab-Measured CD4 Count",
  col="gray", las=1, breaks=60, prob=TRUE)
abline(v=350, col="coral")
lines(density(rd.data$cd4_lab))
```



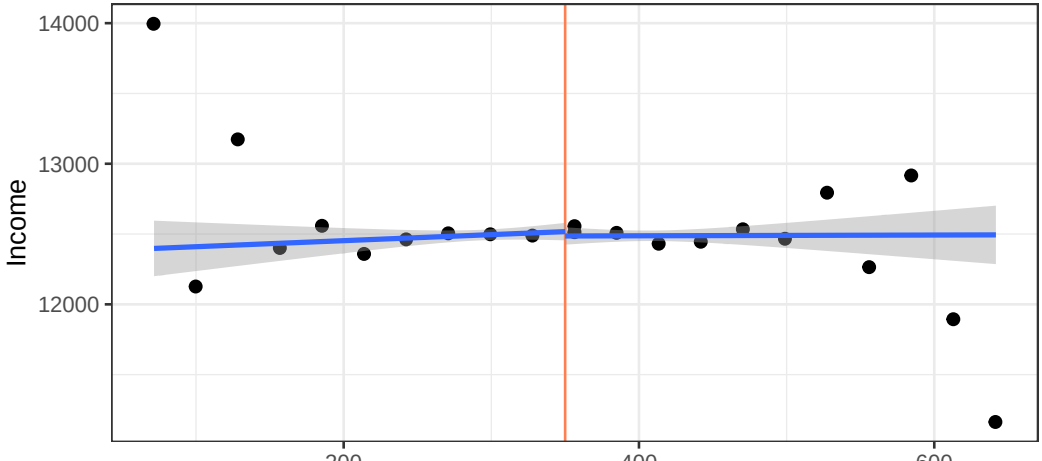
CD4 count: statistical test of bunching?

```r
library(rddensity)
density_test <- rddensity(rd.data$cd4_lab, c = 350)
summary(density_test)
```

```
##
## Manipulation testing using local polynomial density estimation.
##
## Number of obs =       50000
## Model =               unrestricted
## Kernel =              triangular
## BW method =           estimated
## VCE method =          jackknife
##
## c = 350               Left of c           Right of c
## Number of obs         25018               24982
## Eff. Number of obs    11105               8562
## Order est. (p)        2                   2
## Order bias (q)        3                   3
## BW est. (h)           41.606              31.499
##
## Method                T                   P > |T|
## Robust                -0.6356             0.525
##
##
## P-values of binomial tests (H0: p=0.5).
##
## Window Length / 2          <c     >=c    P>|T|
## 0.064                      20      20    1.0000
## 0.128                      39      36    0.8176
## 0.192                      61      56    0.7117
## 0.256                      77      73    0.8066
## 0.319                      95      90    0.7688
```

CD4 count: continuity in observables?

```
ggplot(rd.data, aes(x=cd4_lab, y=income, group=elig)) +
  stat_summary_bin(fun.y='mean', bins=20, size=2, geom='point') +
  geom_vline(xintercept = 350, color = "coral") +
  geom_smooth(method='lm', formula=y~x) +
  labs(x = "Baseline CD4 count", y = "Income") + theme_bw()
```



CD4 count: continuity in observables?

```r
RD.model.incomecheck <- lm(formula = income ~ cd4_lab + elig, data=rd.data)
summary(RD.model.incomecheck)$coefficients
```

```
##                 Estimate  Std. Error   t value  Pr(>|t|)
## (Intercept) 1.239455e+04 129.4454374 95.751138 0.0000000
## cd4_lab     2.289751e-01   0.3147412  0.727503 0.4669213
## elig        3.124580e+01  44.7694452  0.697927 0.4852261
```

CD4 count: causal estimation, full sample

```r
library(ivmodel)
fuzzy_rd_full <- ivmodel(Y = rd.data$Y, D = rd.data$art, Z= rd.data$elig, X = rd.data$cd4_centered)
summary(fuzzy_rd_full)
```

```
##
## Call:
## ivmodel(Y = rd.data$Y, D = rd.data$art, Z = rd.data$elig, X = rd.data$cd4_centered)
## sample size: 50000
## _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
##
## First Stage Regression Result:
##
## F=7648.047, df1=1, df2=49997, p-value is < 2.22e-16
## R-squared=0.1326748,   Adjusted R-squared=0.1326575
## Residual standard error: 0.3957914 on 49998 degrees of freedom
## _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
##
## Coefficients of k-Class Estimators:
##
##              k Estimate Std. Error t value Pr(>|t|)
## OLS     0.0000  19.2554     0.1791  107.51   <2e-16 ***
## Fuller  1.0000  24.5796     0.4960   49.55   <2e-16 ***
## TSLS    1.0000  24.5804     0.4961   49.55   <2e-16 ***
## LIML    1.0000  24.5804     0.4961   49.55   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
##
## Alternative tests for the treatment effect under H_0: beta=0.
##
## Anderson-Rubin test (under F distribution):
## F=2115.47, df1=1, df2=49997, p-value is < 2.22e-16
```

CD4 count: causal estimation, h = 50

```r
rd.data_50 <- subset(rd.data, cd4_centered >= -50 & cd4_centered <= 50)
fuzzy_rd_50 <- ivmodel(Y = rd.data_50 $Y, D = rd.data_50 $art, Z= rd.data_50 $elig, X = rd.data_50$cd4_centered)
summary(fuzzy_rd_50)
```

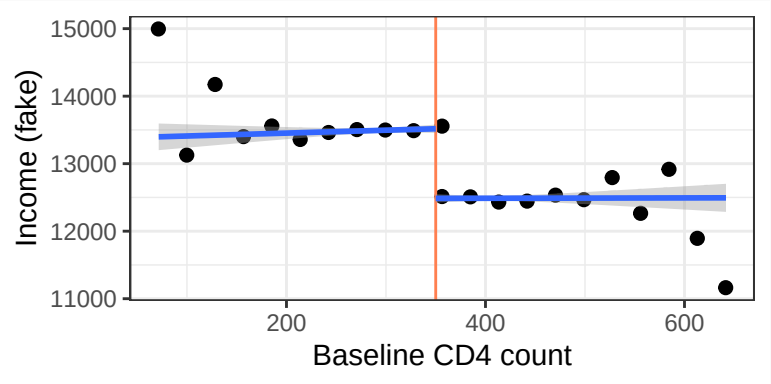
```
##
## Call:
## ivmodel(Y = rd.data_50$Y, D = rd.data_50$art, Z = rd.data_50$elig,
##     X = rd.data_50$cd4_centered)
## sample size: 25934
## _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
##
## First Stage Regression Result:
##
## F=2627.087, df1=1, df2=25931, p-value is < 2.22e-16
## R-squared=0.09199101,   Adjusted R-squared=0.09195599
## Residual standard error: 0.4167074 on 25932 degrees of freedom
## _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
##
## Coefficients of k-Class Estimators:
##
##              k Estimate Std. Error t value Pr(>|t|)
## OLS     0.0000  19.4621     0.2429   80.13   <2e-16 ***
## Fuller  1.0000  25.7550     0.8109   31.76   <2e-16 ***
## TSLS    1.0000  25.7576     0.8111   31.76   <2e-16 ***
## LIML    1.0000  25.7576     0.8111   31.76   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
##
## Alternative tests for the treatment effect under H_0: beta=0.
##
## Anderson-Rubin test (under F distribution):
```

CD4 count: causal estimation, h = 10

```r
rd.data_10 <- subset(rd.data, cd4_centered >= -10 & cd4_centered <= 10)
fuzzy_rd_10 <- ivmodel(Y = rd.data_10 $Y, D = rd.data_10 $art, Z= rd.data_10 $elig, X = rd.data_10$cd4_centered)
summary(fuzzy_rd_10)
```

```
##
## Call:
## ivmodel(Y = rd.data_10$Y, D = rd.data_10$art, Z = rd.data_10$elig,
##     X = rd.data_10$cd4_centered)
## sample size: 5698
## _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
##
## First Stage Regression Result:
##
## F=556.3133, df1=1, df2=5695, p-value is < 2.22e-16
## R-squared=0.08899143,   Adjusted R-squared=0.08883147
## Residual standard error: 0.4273006 on 5696 degrees of freedom
## _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
##
## Coefficients of k-Class Estimators:
##
##              k Estimate Std. Error t value Pr(>|t|)
## OLS     0.0000  19.5994     0.5025   39.00   <2e-16 ***
## Fuller  0.9998  25.1444     1.7009   14.78   <2e-16 ***
## TSLS    1.0000  25.1553     1.7025   14.78   <2e-16 ***
## LIML    1.0000  25.1553     1.7025   14.78   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
##
## Alternative tests for the treatment effect under H_0: beta=0.
##
## Anderson-Rubin test (under F distribution):
```

CD4 count: hypothetical

What if there was a concurrent program that provided a subsidy to PLHIV with a CD4 count below 350? Would you still estimate the CACE?

```
rd.data$income_bonus <- rd.data$income + 1000*rd.data$elig
ggplot(rd.data, aes(x=cd4_lab, y=income_bonus, group=elig)) +
  stat_summary_bin(fun.y='mean', bins=20, size=2, geom='point') +
geom_vline(xintercept = 350, color = "coral") + geom_smooth(method='lm', formula=y~x) +
  labs(x = "Baseline CD4 count", y = "Income (fake)") + theme_bw()
```